```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv("train.csv")
df.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Tick
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 211
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 175
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/(31012
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	1138
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	3734

Next steps: Generate code with df | New interactive sheet

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   PassengerId  891 non-null    int64
 1   Survived     891 non-null    int64
 2   Pclass       891 non-null    int64
 3   Name         891 non-null    object
 4   Sex          891 non-null    object
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64
 7   Parch        891 non-null    int64
 8   Ticket       891 non-null    object
 9   Fare         891 non-null    float64
```

```
 10  Cabin        204 non-null    object
 11  Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

```
df.describe()
```

	PassengerId	Survived	Pclass	Age	SibSp	Parch
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000

```
df.isnull().sum()
```

	0
PassengerId	0
Survived	0
Pclass	0
Name	0
Sex	0
Age	177
SibSp	0
Parch	0
Ticket	0
Fare	0
Cabin	687
Embarked	2

dtype: int64

```
df.shape
```

```
(891, 12)
```

```
df['Survived'].value_counts()
```

Survived	count
0	549
1	342

dtype: int64

```
df['Sex'].value_counts()
```

Sex	count
male	577
female	314

dtype: int64

```
df['Pclass'].value_counts()
```

Pclass	count
3	491
1	216
2	184

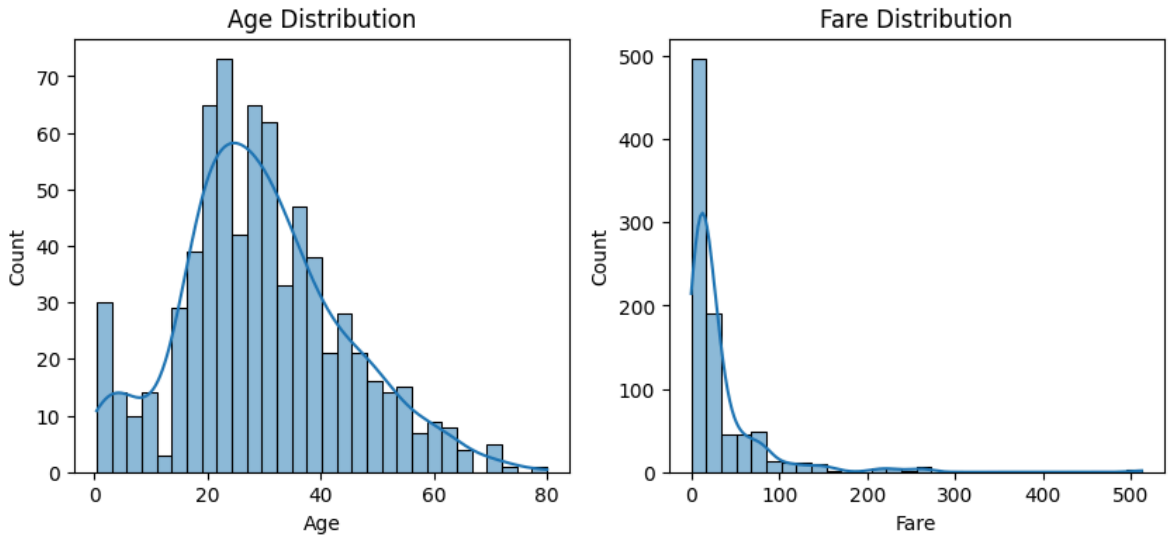
dtype: int64

```
plt.figure(figsize=(10,4))

plt.subplot(1,2,1)
sns.histplot(df['Age'].dropna(), bins=30, kde=True)
plt.title('Age Distribution')

plt.subplot(1,2,2)
sns.histplot(df['Fare'], bins=30, kde=True)
plt.title('Fare Distribution')

plt.show()
```

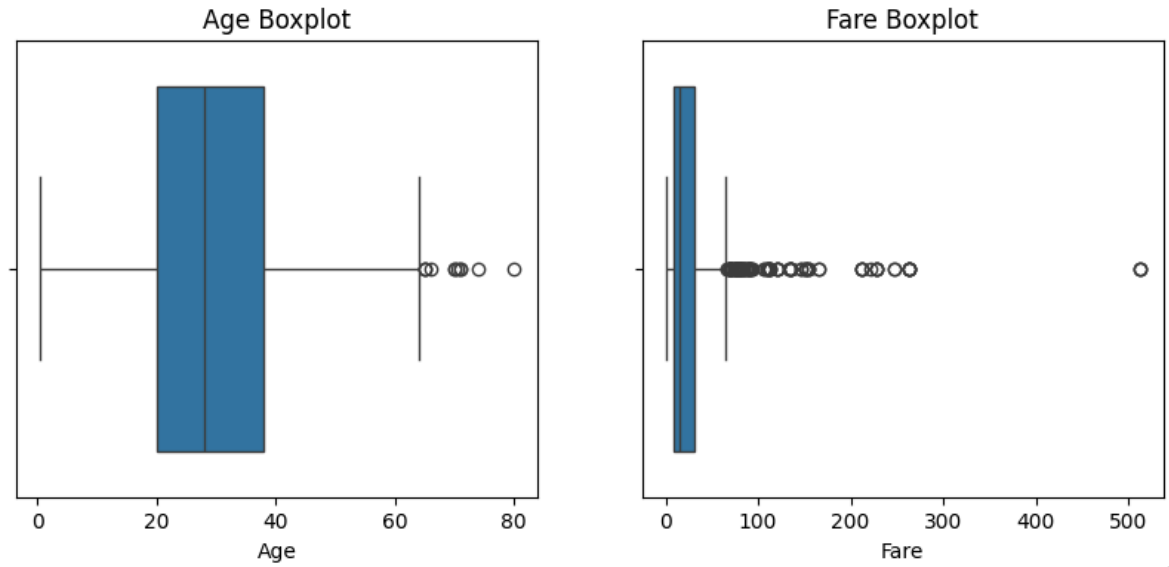



```
plt.figure(figsize=(10,4))

plt.subplot(1,2,1)
sns.boxplot(x=df['Age'])
plt.title('Age Boxplot')

plt.subplot(1,2,2)
sns.boxplot(x=df['Fare'])
plt.title('Fare Boxplot')

plt.show()
```



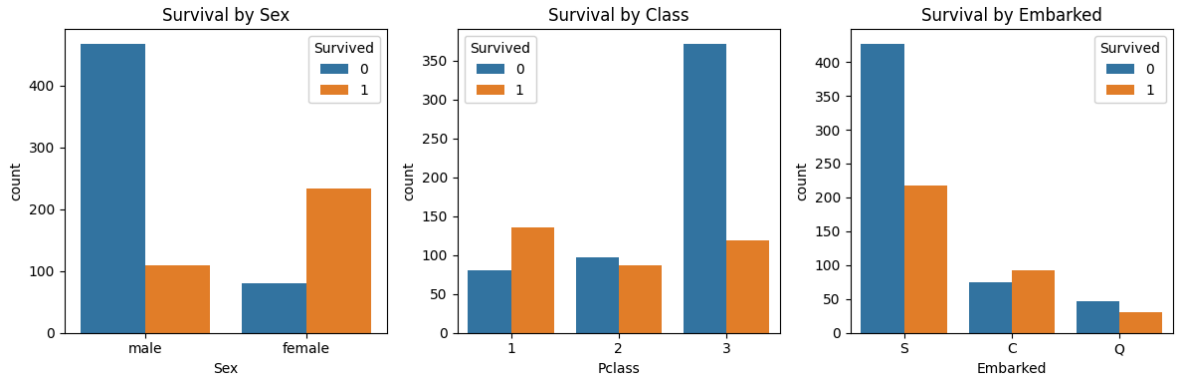

```
plt.figure(figsize=(12,4))

plt.subplot(1,3,1)
sns.countplot(x='Sex', hue='Survived', data=df)
plt.title('Survival by Sex')

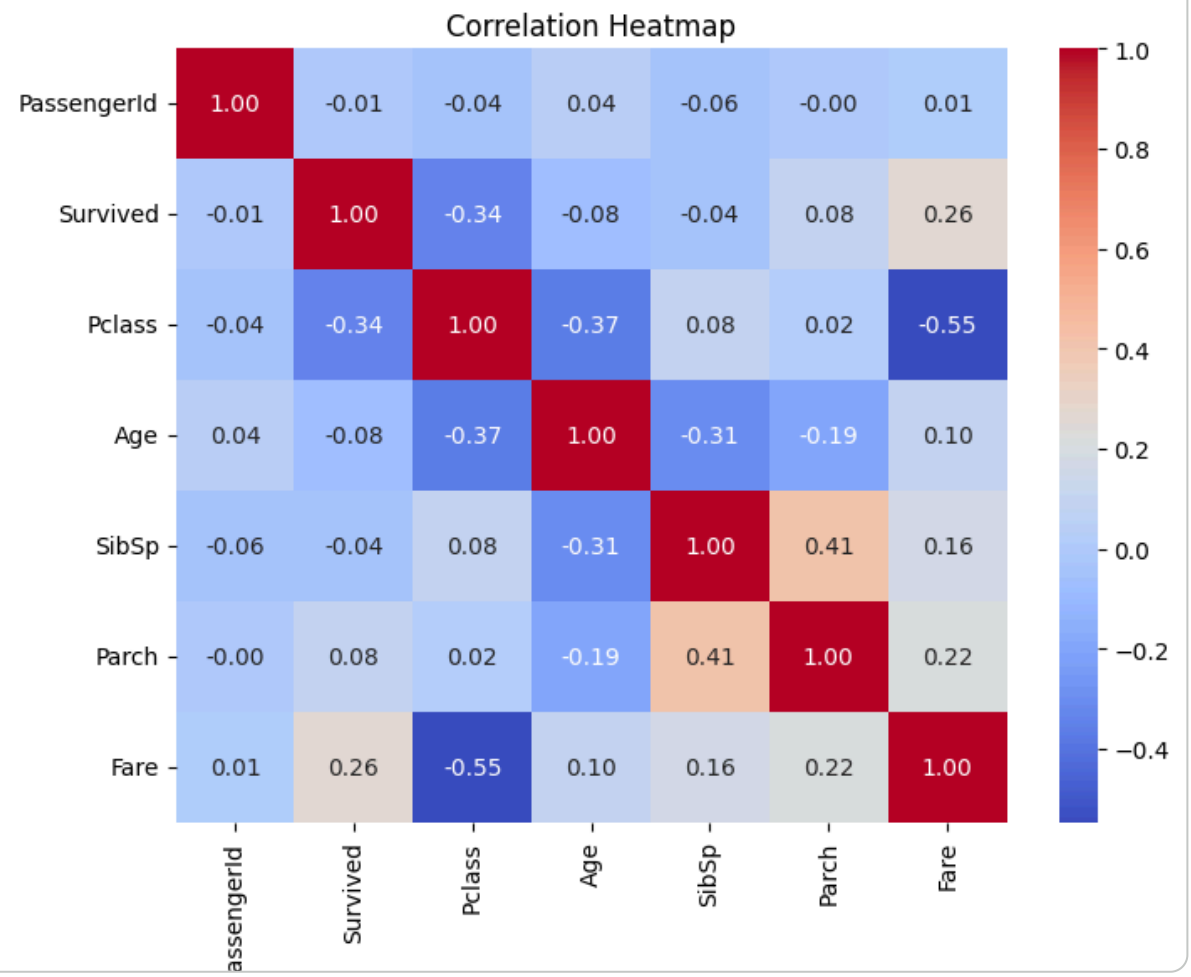
plt.subplot(1,3,2)
sns.countplot(x='Pclass', hue='Survived', data=df)
plt.title('Survival by Class')
```

```
plt.subplot(1,3,3)
sns.countplot(x='Embarked', hue='Survived', data=df)
plt.title('Survival by Embarked')

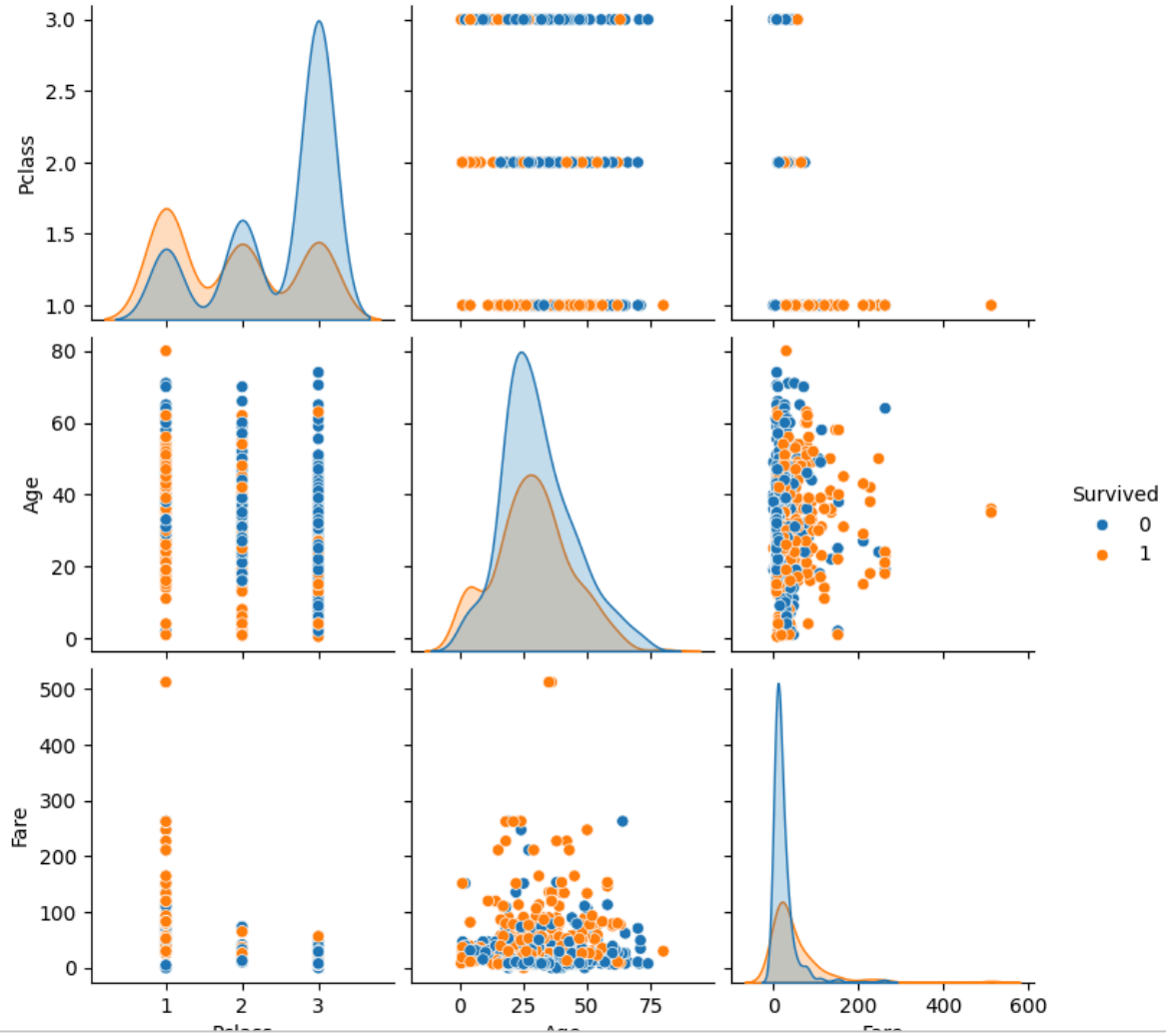
plt.tight_layout()
plt.show()
```




```
plt.figure(figsize=(8,6))
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap='coolwarm', fr
plt.title('Correlation Heatmap')
plt.show()
```




```
sns.pairplot(df[['Survived','Pclass','Age','Fare']].dropna(), hue='Survived'
plt.show()
```




```
print("""
SUMMARY OF FINDINGS

1. Dataset: 891 passengers, 12 columns. Age and Cabin la null values ir
2. Survival rate: ~62% iranthuttaanga, ~38% uyir thapinaanga.
3. Gender: Female survival rate (~74%) romba adhigam, male (~19%) kammi
4. Class: 1st class survival adhigam, 3rd class la death adhigam.
5. Age: Perumbalanor 20-40 vayasu la. Kuzhandhaigal (0-5) survival nal
6. Fare: Right-skewed, extreme outliers (200-500). Fare adhigam kudutha
7. Correlation: Survived vs Pclass (-0.34), Survived vs Fare (0.26). Pc
8. Anomalies: Fare la extreme outliers, Age la 65+ outliers.
9. Conclusion: Gender, passenger class, and fare than survival ah romba
""")
```

```
SUMMARY OF FINDINGS

1. Dataset: 891 passengers, 12 columns. Age and Cabin la null values irukku.
2. Survival rate: ~62% iranthuttaanga, ~38% uyir thapinaanga.
3. Gender: Female survival rate (~74%) romba adhigam, male (~19%) kammi.
4. Class: 1st class survival adhigam, 3rd class la death adhigam.
5. Age: Perumbalanor 20-40 vayasu la. Kuzhandhaigal (0-5) survival nalla irun
6. Fare: Right-skewed, extreme outliers (200-500). Fare adhigam kuduthavanga
7. Correlation: Survived vs Pclass (-0.34), Survived vs Fare (0.26). Pclass v
8. Anomalies: Fare la extreme outliers, Age la 65+ outliers.
9. Conclusion: Gender, passenger class, and fare than survival ah romba affec
```